NIELIT bioAI CONCLAVE 2026

Track 2 – Genomics, Multi-Omics & Precision Health

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What Near-Perfect Accuracy Does and Does Not Show

Leakage, Equivalence, Conformal Reliability and Data Efficiency on the UCI TCGA Pan-Cancer RNA-Seq Dataset

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1. Introduction: Motivation

- Tissue of origin guides cancer treatment, notably for unknown primaries [5].
- Expression profiles carry a strong cell-of-origin signal [2].
- On the UCI TCGA subset, 98–100% accuracy is routinely reported [1], [7].
- Many studies select genes on all data and use no test [3], [6]: how much does it matter here?

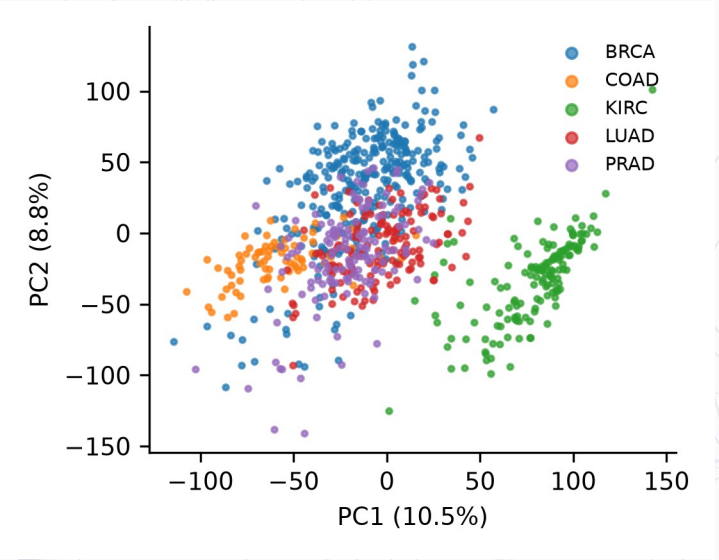


Fig. 1. PCA of all expressed genes, 801 tumours, five types (visualisation only) [1].

1. Introduction: The Dataset

UCI dataset 401 · TCGA RNA-Seq (HiSeq) · 801 tumours × 20,531 genes [1]

Classes

BRCA 300, KIRC 146, LUAD 141, PRAD 136, COAD 78.

Values

Already log-scaled (0–20.8); 267 genes constant; no missing values.

Gene names

Dummy names gene_0 … gene_20530; no gene symbols released.

Not included

Clinical data, batch or site information, normal tissue labels.

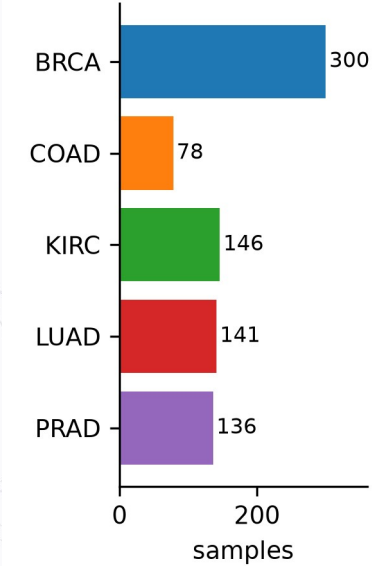


Fig. 2. Samples per tumour type; COAD is the smallest class.

1. Introduction: Research Questions

RQ1 How much does selecting genes on all data change accuracy here?

RQ2 Do any of seven classifiers differ, and which are equivalent?

RQ3 Calibrated, abstaining when unsure? How much data is needed?

Common practice

Select genes on all 801 samples, then split.

Single split; winner declared without a test.

This study

In-fold selection; leaky variant audited.

10×5 repeated CV, corrected CIs.

All 21 pairs, Holm, TOST, Bayesian.

2. Methodology

Macro-F1 primary · identical splits for all models · no tuning

Pipeline

Per training fold: drop constant genes, ANOVA top-100, z-score.

Models

Softmax, ridge, linear SVM, naive Bayes, RF, LightGBM, MLP; majority baseline.

Evaluation

10×5 repeated stratified CV (50 folds); accuracy, macro-F1, ROC-AUC.

Statistics

Corrected CIs [4]; all 21 pairs (Holm); TOST ±1 pt; Bayesian ROPE.

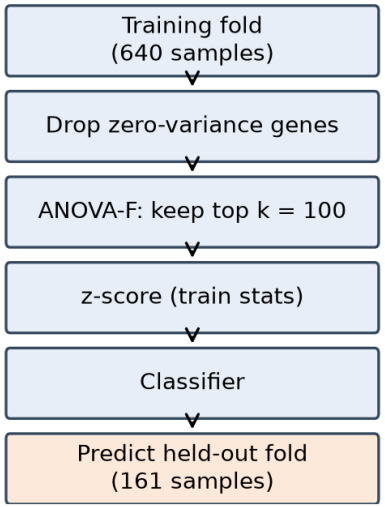


Fig. 3. Steps refitted inside every training fold.

3. Results: Leakage Audit (softmax accuracy)

Labels	n	Genes k	Leaky	In-fold
Real	801	100	99.50%	99.48%
Real	801	10	95.22%	92.32%
Permuted	801	100	35.9%	25.3%
Real	60	100	99.9%	99.3%
Permuted	60	100	68.2%	19.9%

Leaky: gene filter, ANOVA selection and scaling fitted on all samples before CV. Chance with permuted labels: 24% (n = 801), 20% (n = 60).

- Real labels, n = 801: leakage changes accuracy ≤ 0.29 pt for k ≥ 20.
- The bias shows only on permuted labels, strongly at n = 60 [3]: it hides behind the ceiling.

3. Results: Leakage-Free Performance

Model	Macro-F1, %	Errors / 801
Majority baseline	10.90	501.0
Softmax	99.48	4.2
Ridge	99.73	2.6
Linear SVM	99.81	1.5
Naive Bayes	98.90	8.2
Random forest	99.53	3.6
LightGBM	99.40	4.5
MLP (64)	99.44	4.3

Mean over 50 folds, top-100 genes. Errors: misclassified samples per repeat. ROC-AUC ≥ 0.9944 for all seven.

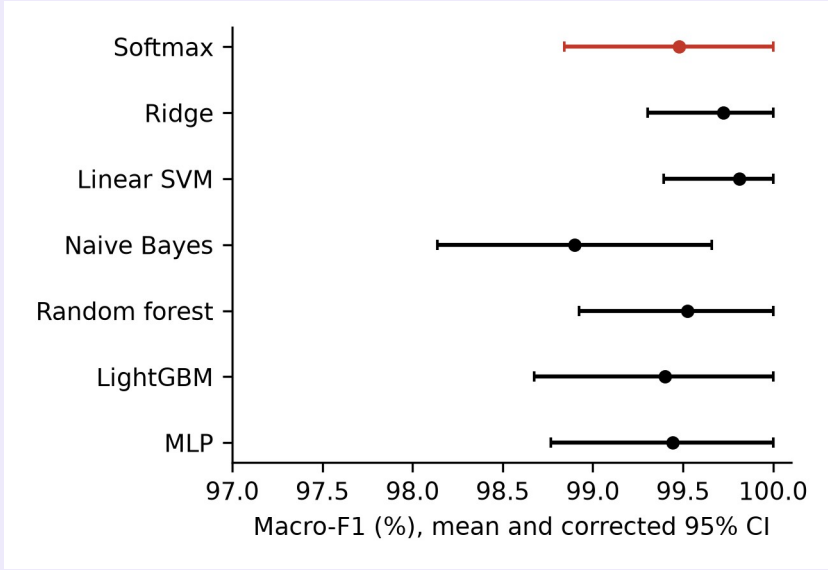


Fig. 4. Macro-F1 with corrected 95% CI; red: softmax regression.

3. Results: All Pairwise Comparisons

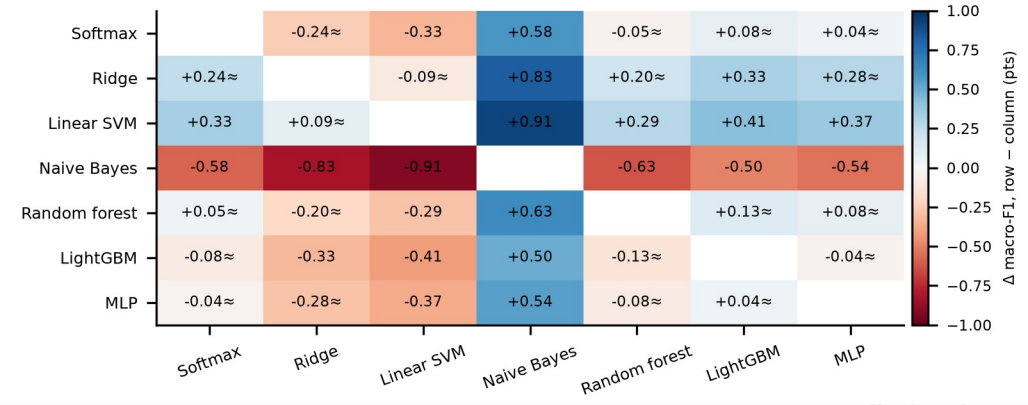


Fig. 5. Δ macro-F1 for all pairs (row - column); ≈ : equivalent within ±1 pt (TOST, Holm).

All 21 — No pair significant; smallest Holm p = 0.66 (Linear SVM vs Naive Bayes).

TOST — 10 of 15 pairs without naive Bayes equivalent within ±1 pt.

Power — Detectable difference (80% power): 0.5–1.4 pts.

Meaning — No winner; most pairs equivalent; naive Bayes probably slightly worse.

3. Results: Errors and Reliability

Errors (softmax, 10 repeats)

- 42 errors in 8,010 predictions.
- Most common: LUAD → BRCA (27), COAD → LUAD (8).

Calibration and conformal sets

- ECE 0.47% vs 0.14% if calibrated: small, probably real.
- 99%: accepted error 0.18% (all: 0.51%); threshold 0.26%.
- Hard LUAD flagged 18/30; Mondrian 99%: needs ≥ 99/class.

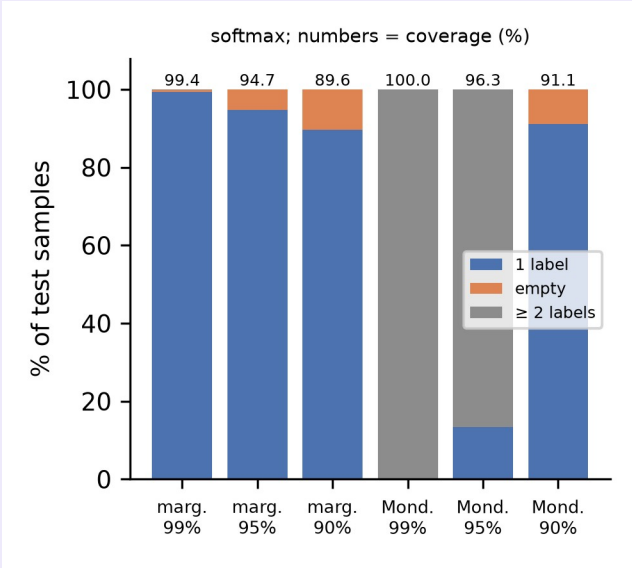


Fig. 6. Split-conformal set types, nested in 10×5 CV.

4. Results: How Much Data?

Linear models: 99.4% macro-F1 (±0.2) with 10 tumours per class.

Learning — 3 per class: macro-F1 softmax 98.1 ± 0.7%; naive Bayes 71.4%.

Leakage — Bias on noise: 65 ± 5 pts at n = 30, 14 ± 2 at n = 390 [9].

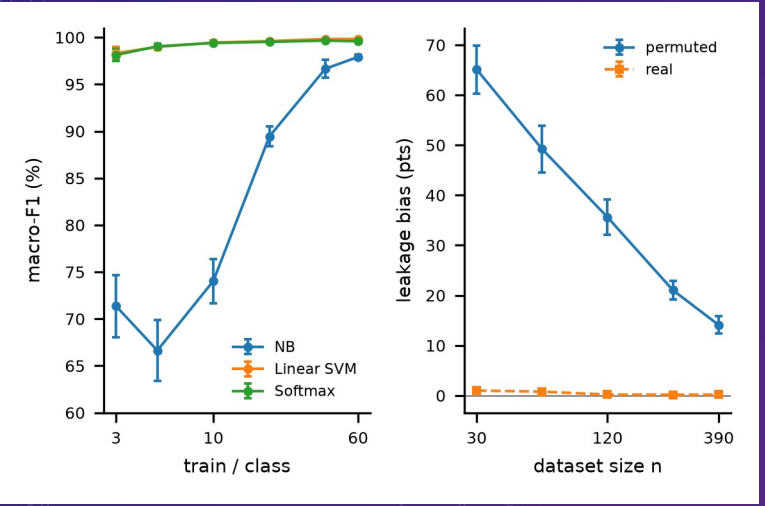


Fig. 7. Learning curves; leakage bias vs n (95% CI).

4. Results: Selection Stability



Fig. 8. How often each gene was in the top 100 across 50 training folds.

Ever — 133 genes entered the top 100 in some fold.

Always — 76 genes in all 50 folds; 85 in ≥ 90%.

Overlap — Jaccard 0.87; Nogueira Φ = 0.93.

vs [8] — Sirohi: RF panel, Jaccard 0.32. Stability is method-dependent.

5. Discussion: Limitations

What remains valid

Leakage size, equivalence and stability on this dataset, within TCGA, five tissues.

Data

- n = 801, single source, five types; no metadata.
- Global profile statistics differ by class.

Methods

- No external cohort or metastatic samples.
- Tests approximate for bounded fold scores.
- Audit covers selection, not SMOTE or PCA.

6. Conclusion: Takeaways and Future Work

Takeaways

- At n = 801, leaky selection barely changes accuracy.
- No model significantly better; 10/15 pairs equivalent (±1 pt).
- 99% conformal abstention: accepted error 0.18% (threshold-like).
- Near-100% here proves neither a clean pipeline nor a better model.

Future work

- Map gene indices to symbols via the original annotation.
- External, non-TCGA validation cohorts.
- The full 33-type TCGA problem.
- Metastatic and unknown-primary samples.

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THANK YOU

FOR YOUR ATTENTION

Questions?

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